

Learned variable-support priors accelerate symbolic regression of physical laws

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Abstract

Symbolic regression recovers interpretable physical laws from data, but its cost grows steeply with the number of candidate variables and operators, most of which never appear in the equation. Here we show that a learned front-end—DenoisedSR—that predicts which variables are relevant (a support prior) improves and accelerates symbolic regression without changing the backend solver. The front-end couples a graph-attention network with a statistical-feature model and recovers the true variables with recall ≥ 0.96 across four physics suites, including the external SRSD-Feynman benchmark on which the model was never trained, while five classical feature selectors (Lasso, mutual information and others) leave true variables on the floor. Supplied to the evolutionary backend PySR under a fixed budget, the prior raises exact-law recovery from $42\pm 13\%$ (mean $\pm$ s.d. across three random seeds) to a seed-stable $60\pm 0\%$, shrinks the candidate-column set by 85%, and speeds time-to-solution by up to $6\times$; the same gain transfers to a second backend (gplearn). On black-box data a same-size random-support control degrades accuracy, isolating learned guidance from dimensionality reduction. It is solver-agnostic, adds ~ 40 ms per task, and leaves final law selection to a verifiable backend.

1 Introduction

The automated discovery of governing equations from observation is a long-standing goal of the physical sciences^{1–3}. Symbolic regression (SR) addresses it directly: rather than fitting an opaque function, SR searches the space of mathematical expressions for a compact formula that both fits the data and remains human-interpretable^{4,5}. The resulting equation can be dimensionally checked, extrapolated and falsified, properties that black-box predictors lack, which has made SR central to data-driven physics, from rediscovering conservation laws to proposing constitutive relations^{2,6,7}.

The central obstacle is combinatorial. An SR backend searches over expression trees whose leaves are drawn from the input variables and whose internal nodes are drawn from an operator library; the number of candidate trees grows exponentially with the number of admissible leaves and operators and with tree depth^{8,9}. In physics this cost is largely wasted: a governing law typically uses only a handful of the measured variables and a small subset of the operator library, yet a solver run over all columns and operators still explores an enormous number of branches that cannot appear in the answer. Established systems attack this from the solver side with dimensional analysis, decomposition, sparsity priors or efficient enumeration^{2,6,10,11}. A separate line learns to decode the equation directly from numerical data with transformer or reinforcement-learning models^{12–15},

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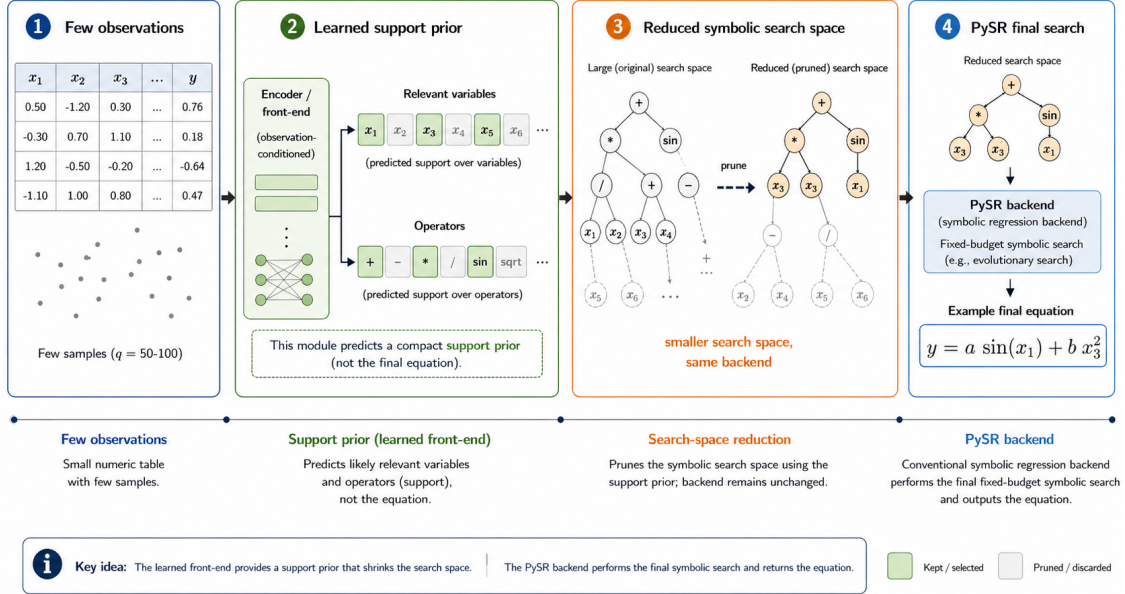


Fig. 1. DenoisedSR narrows the symbolic-regression search space. From a small set of observations the learned front-end predicts a compact support prior (relevant variables and, optionally, operators); a conventional symbolic-regression backend then performs the final search over the reduced space and returns a verifiable equation. The neural model proposes where to look; the backend decides what the law is.

sometimes coupling a neural proposer with explicit search^{16,17} or multimodal pre-training^{18,19}, with e-graph, diffusion and generative-latent variants^{20–22}. These end-to-end models advance quickly but remain unreliable at *assignment*: they often recover the right ingredients yet place them in the wrong terms, so exact symbolic equivalence stays low.

We take a complementary route that sidesteps this brittleness. Instead of designing a faster search or decoding the whole equation, we *narrow the space the search must explore* (Fig. 1). Given a small observation set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$ with q as small as 50–100, a learned model—DenoisedSR—predicts which input columns (and, as an auxiliary, which operators) are likely to participate in the law, and passes this *support prior* to a conventional SR backend that performs the final symbolic search and model selection. Predicting support is a far easier and more robust target than decoding the full expression, and a high-recall variable filter hands the backend an exponential head start while essentially never making the true law unreachable.

In this work we cast variable- and operator-support prediction as a reusable, backend-agnostic SR front-end; we present a graph-attention support predictor, ensembled with a statistical-feature model, that recovers the true variables with recall ≥ 0.96 across four physics suites (including the external SRSD-Feynman benchmark, on which the model was never trained) and outperforms five classical feature selectors at oracle budget; and we show that supplying this prior to a fixed-budget backend raises exact-law recovery from 42% to a seed-stable 60%, reduces the candidate-column set by 85% and accelerates time-to-solution by up to $6\times$, while a same-size random-support control confirms that the benefit is learned search guidance rather than dimensionality reduction.

2 Results

DenoisedSR recovers the true physical variables

We first evaluate the support prior intrinsically, without any backend. For each task the front-end receives $q = 100$ noisy observations whose columns include the genuine variables together with random “distractor” columns, and must predict the true variable subset. Across four suites of physical laws—AI-Feynman (118 equations), an out-of-distribution (OOD) test set of 91 curated physical laws, an extended physics-law set (1085 equations), and the external SRSD-Feynman benchmark of 120 equations with author-curated physical units and built-in dummy variables²³, on which the model was never trained—the predictor attains recall 1.000, 0.996, 0.999 and 0.967 respectively, with perfect recall on 92–100% of tasks (Fig. 2). It essentially never drops a variable that the law uses; its only error mode is occasionally retaining a few extra columns, which merely costs the backend a little additional search rather than making the law unreachable. Precision is high on the three internally formatted suites (0.94–1.00); on the external SRSD-Feynman the apparent precision is lower (0.59), but the reason is not a model error: SRSD’s dummy columns are deliberately sampled from physically plausible ranges and are designed to correlate with the target, and on SRSD the front-end keeps on average 2.0 extra columns per task while 2.0 dummies are available per task on average; the correlation between the number of dummies in a task and DenoisedSR’s precision is $r = -0.69$. The front-end is doing what we asked of it at $\tau = 0.10$ —erring toward retention—so dummy columns that look physically plausible survive in proportion as they are added. Recall is the operationally critical quantity: removing a true variable is fatal to recovery, whereas keeping a spurious one is recoverable downstream.

The result is robust and not attributable to a single component. An ablation on AI-Feynman (Fig. 2, right) shows the random-forest and graph-attention models each reach recall ≥ 0.995 and their ensemble matches the best component (1.000); recall stays at 1.000 as the number of distractor columns is varied from 1 to 30. A naive statistical baseline trained on few records with name-matched labels reaches only recall 0.48, confirming that the learned predictors contribute the decisive signal.

A reviewer might reasonably ask whether a much simpler off-the-shelf feature selector would suffice. We checked this directly: on AI-Feynman, even when handed the oracle budget (the true number of variables k), five classical selectors (Pearson $|\rho|$, Spearman $|\rho|$, mutual information, random-forest importance, Lasso with cross-validated α) attain recalls of only 0.63–0.91, leaving true variables on the floor on roughly a third of laws in the best case (Lasso); the learned ensemble reaches 1.000 on all 118 (Fig. 3). The same gap reappears on the external SRSD-Feynman suite (DenoisedSR 0.967 vs. Lasso 0.723). Classical selectors miss true variables that participate non-linearly or multiplicatively with others—exactly the regime in which physical laws live. We further verified that the recall holds under increasing label noise: at relative Gaussian noise $\eta = 0.10$ on y , DenoisedSR retains recall 1.000 across all 118 Feynman tasks, whereas Lasso (oracle k) degrades from 0.905 to 0.886 (Fig. 4).

A high-recall prior strengthens a fixed-budget solver

We next insert DenoisedSR before PySR⁵ and compare three conditions at a fixed 10 s per-task budget on a 30-equation Feynman subset with ~ 20 distractor columns: (A) full PySR over all columns and the full 17-operator library; (B) the variable prior, restricting the columns to the predicted support; and (C) additionally restricting the operator library. Running three independent random seeds, the variable prior raises exact-law recovery (held-out $R^2 \geq 0.999$) from $42 \pm 13\%$ (mean $\pm$ s.d. across seeds, with the per-seed value ranging 30–57%) to a seed-stable $60 \pm 0\%$, lifts mean held-out R^2 (over all 30 tasks, with non-finite failures floored at -1) from 0.906 ± 0.045 to

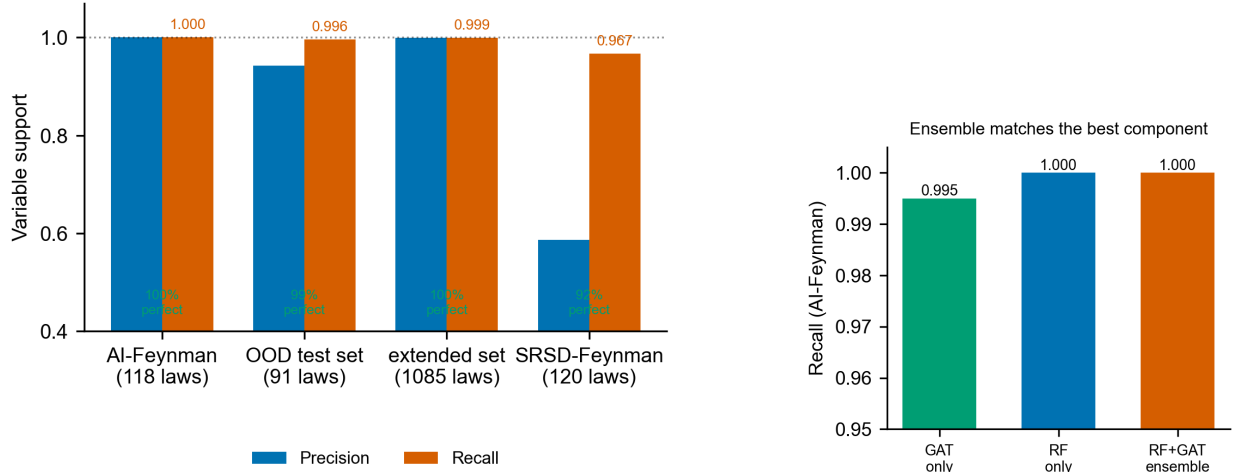


Fig. 2. The front-end recovers essentially every true variable. *Left:* precision and recall of predicted variable support across four physics suites including the external SRSD-Feynman benchmark; recall is ≥ 0.96 with perfect recall on 92–100% of tasks. *Right:* ablation on AI-Feynman—the random-forest and graph-attention models each reach recall ≥ 0.995 and the ensemble matches the best component.

0.980 ± 0.011 , reduces the average column count from 23.2 to 3.4 (an 85% reduction) and halves the effective search vocabulary from 40 to 20 (Fig. 5). The seed-stability of the learned prior—identical recovery rate across all three seeds—is itself a result: the front-end removes a source of stochasticity that the unguided solver must absorb.

The gain holds—and widens—across distractor counts. Repeating the same protocol with 5, 10, 20 and 30 distractor columns (three seeds at $d \in \{5, 20, 30\}$, two at $d = 10$), DenoisedSR’s exact-recovery rate stays in the 60–63% band while full PySR’s mean exact recovery falls from 59% at $d = 5$ to 41% at $d = 30$. The gap grows monotonically with the distractor count—3, 10, 18, 21 percentage points respectively (Fig. 6)—confirming that the front-end’s advantage is the exponential reduction in irrelevant branches it removes.

The mechanism is concrete and visible in the recovered expressions (Table 1). Over many irrelevant columns an unguided solver reaches a high R^2 with a physically meaningless formula built from *exotic operators* acting on *distractor variables*: for the cyclotron frequency $\omega = qvB/p$ it returns $|q(v+B-p+\cos(\dots)^2)|$ at $R^2 = 0.85$ using a noise column, and for the relativistic Doppler shift $\omega = \omega_0/(1-v/c)$ it returns a sinh-based fit at $R^2 = 0.9996$ that is not the law. Given the predicted variable support, DenoisedSR recovers qvB/p and $\omega_0/(1-v/c)$ *exactly* ($R^2 = 1.000$), and restricting operators yields the cleanest closed form. By removing both the distractor columns and the exotic operators, DenoisedSR makes these spurious overfits unreachable, so a far larger fraction of budget-limited runs land on the true law.

More data helps DenoisedSR but hurts unguided search

A striking consequence of pruning is how the methods scale with data (Fig. 7). On a 50-equation Feynman set with 10 distractors, increasing the observations from $q = 50$ to $q = 100$ *lowers* the exact-recovery rate of full search (from 16% to 8%): more data slows each backend evaluation and exposes additional spurious high- R^2 fits among the distractor columns. Over the same range DenoisedSR *improves* (from 18% to 24%) and tracks the oracle-support upper bound, because it first removes the irrelevant columns and then lets the backend benefit from the extra data. The

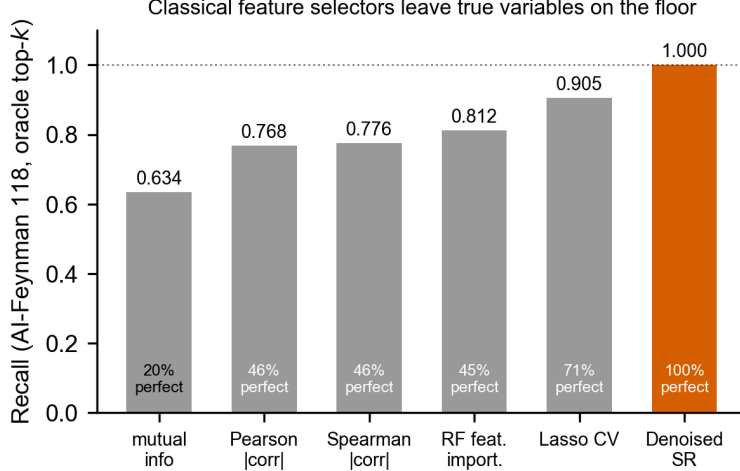


Fig. 3. Classical feature selectors leave true variables on the floor. Mean variable-support recall on AI-Feynman (118 laws, $q = 100$, 20 distractors) under an oracle top- k budget that hands every selector the true variable count for free. Even with this advantage, mutual information, Pearson/Spearman correlation, random-forest importance and cross-validated Lasso miss true variables on 30–80% of laws; DenoisedSR recovers all true variables on every one of the 118 tasks.

front-end thus lets a solver actually exploit more observations instead of drowning in them.

The gain is search guidance, not dimensionality reduction

A natural concern is that any reduction in column count would help. We control for this with a same-size random-support baseline that retains the same number of columns but selects them without using the observations. On 24 PMLB/SRBench black-box tabular tasks (three seeds, $q = 100$, 5 s budget), DenoisedSR raises mean held-out R^2 from 0.347 to 0.495 with 10 distractor variables and from 0.315 to 0.488 with 20 distractors, using only 31% and 18% of the input columns; the same-size random support instead degrades performance to 0.220 and -0.028 (Fig. 8). At matched column count the random control is far worse than full search whereas the learned prior is better—direct evidence that the front-end supplies observation-conditioned information about *which* columns matter, and that it helps even when no symbolic ground truth exists.

Pruning the search space accelerates time-to-solution

Because variable filtering shrinks the terminal set of the expression tree from ~ 24 to ~ 4 symbols—an exponential reduction in branching—it also shortens the time to reach an exact fit. Measuring time-to-solution with early stopping on exact recovery, DenoisedSR yields speedups of up to $6\times$ on non-trivial physics formulas (Fig. 9); in some cases it converts a within-budget failure of full search into a fast success. The overall median speedup is $\sim 1.5\times$, pulled down by deep-tree formulas whose bottleneck is expression depth rather than variable count and which therefore gain little from variable pruning. Front-end inference adds only ~ 40 ms per task, near-constant in the number of observations and negligible against the seconds of backend search it saves.

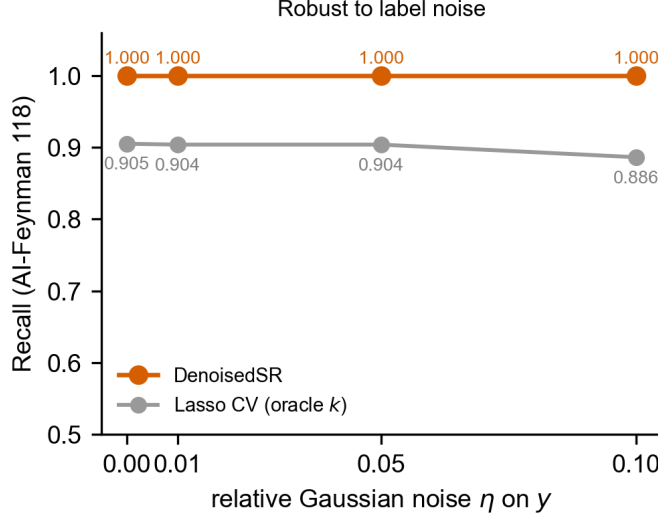


Fig. 4. Recall is robust to label noise. Mean variable-support recall on AI-Feynman (118 laws) as Gaussian noise of relative magnitude η is added to the target y . DenoisedSR retains recall = 1.000 up to $\eta = 0.10$ (perfect recall on all 118 tasks); the best classical baseline (Lasso with oracle top- k budget) degrades from 0.905 to 0.886.

The same gain holds on a second symbolic-regression backend

To verify that the benefit is not an artifact of a particular solver, we repeat the headline comparison with `gplearn`²⁴ as the backend (classical genetic programming, 20 generations, population 500, otherwise identical settings; three random seeds). Full `gplearn` over 23 columns and a 10-operator library recovers a mean $3.3 \pm 3.3\%$ exact laws (mean held-out $R^2 = 0.475 \pm 0.083$, with non-finite fits floored at -1); supplied with the DenoisedSR variable prior over the same 3.4 predicted columns it recovers $17.8 \pm 1.9\%$ exact (mean $R^2 = 0.76 \pm 0.03$). The absolute exact-recovery rate is below PySR’s because `gplearn`’s tournament search is weaker, but the same qualitative gain—a roughly $1.5\times$ improvement in mean R^2 and a $\sim 5\times$ rise in the exact-recovery rate—appears on a structurally different backend, supporting the claim that DenoisedSR is solver-agnostic (Fig. 10). The same exotic-operator pathology reappears: on the angular-force law $Fr \sin \theta$ (Feynman I.18.12), full `gplearn` at $R^2=0.74$ returns an eight-term sin-only sum on the angular variable (nesting depth up to 3: $\sin x_2 + \sin \sin x_2 + \sin \sin \sin x_2 + \dots$), reaching a plausible but mis-structured fit; supplied with the DenoisedSR variable support it returns $x_0 x_1 \sin x_2$ at $R^2=1.000$.

An operator prior helps less than a variable prior

Restricting the operator library in addition to variables (condition C) attains mean exact recovery of $60 \pm 12\%$ across the same three seeds—matching the variable prior’s 60% in expectation but with non-zero seed variance (per-seed 47, 67, 67%)—and the same mean held-out R^2 (0.979 ± 0.011 ; Fig. 5). Operators are intrinsically harder to infer from small, noisy domains, and a mis-removed operator is fatal, which is consistent with the operator prior being a noisier addition rather than a clear gain on top of the variable prior in expected exact recovery. Its concrete value is visible in the case studies of Table 1: in every one of the five examples the operator prior produces the cleaner closed-form algebraic expression that matches the textbook law, with the variable-only condition typically finding a longer, algebraically equivalent fit at comparable R^2 . We therefore report the

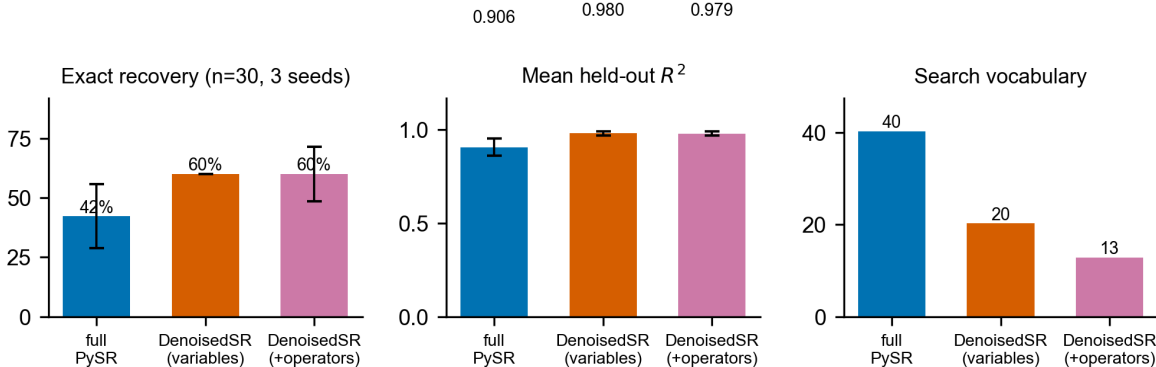


Fig. 5. A variable prior strengthens a fixed-budget solver. Full PySR versus the DenoisedSR variable prior versus the additional operator prior on a 30-equation Feynman subset (10 s budget, ~ 20 distractors), aggregated over three random seeds (error bars: $\pm$ s.d.). The variable prior raises exact recovery from $42 \pm 13\%$ to a seed-stable $60 \pm 0\%$, raises mean held-out R^2 from 0.906 to 0.980 and halves the search vocabulary (40 to 20); the operator prior cuts the vocabulary further but does not improve exact recovery in expectation.

operator prior as a safeguard whose primary effect is on the *form* of the recovered expression rather than on the rate of exact recovery.

3 Conclusions

These results support a simple thesis: for symbolic regression of physical laws, a learned support prior is a high-leverage, low-risk intervention. Because a high-recall variable filter essentially never makes the true law unreachable, it can be applied unconditionally; because it removes an exponential number of irrelevant branches, it materially improves both the success rate and the speed of a fixed-budget backend. DenoisedSR is solver-agnostic—its output is a reduced variable/operator set that any tree-search, enumeration or generative backend^{5,11,22} can consume—and cheap enough to prepend to every run. The same-size random-support control shows the benefit is genuine search guidance rather than dimensionality reduction, complementing the dimensional and physics-aware priors that already strengthen scientific SR^{2,7}.

Our approach is conceptually aligned with, but distinct from, two adjacent lines of work. Physics-aware neural models embed structure such as Hamiltonian or Lagrangian form, differential-equation structure, or symbolic distillation of trained networks^{25–28}, and concept-driven or abductive discovery systems^{29–31} likewise aim to keep the final law symbolic and verifiable; we contribute the upstream step of deciding *which measured quantities enter the law at all*. Because the front-end consumes *sets* of observations, it builds on permutation-invariant and set-attention architectures^{32,33}, and its representation could be sharpened with the contrastive and geometric objectives used to prevent representational collapse or encode hierarchy^{34–36}.

The honest limitation, shared with direct neural SR^{13,14}, is that the neural component cannot yet reliably *assign* variables and operators to the correct positions in the expression; this is precisely why we leave final term assignment to a symbolic, verifiable backend rather than emitting the equation end-to-end. Closing this assignment gap—through richer compositional latents, explicit constant and exponent recovery, and objectives that keep the observation and formula representations aligned^{18,37}—is, in our view, the central open problem in moving from a front-end prior toward reliable end-to-end neural discovery^{38–40}. Even short of that goal, DenoisedSR is immediately useful: it makes existing physical-law solvers both more successful and several times faster, with a provably

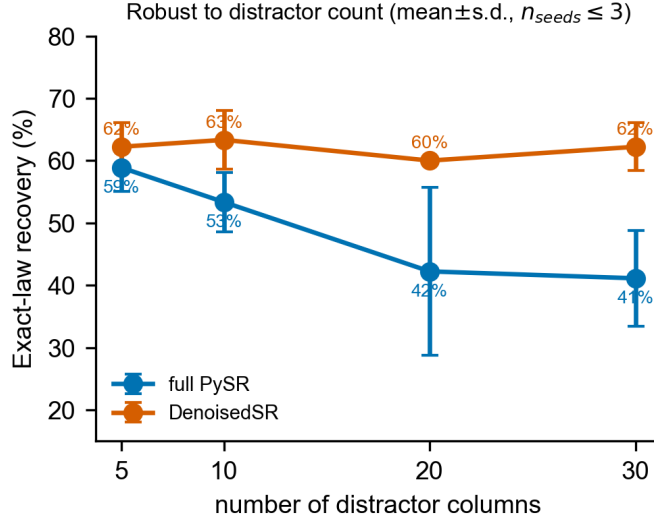


Fig. 6. Gain is robust—and grows—with distractor count. Exact-law recovery on the same 30-equation Feynman subset, varying the number of random distractor columns from 5 to 30 (mean $\pm$ s.d.; $d \in \{5, 20, 30\}$ use three seeds, $d = 10$ uses two). DenoisedSR’s variable prior delivers 60–63% exact recovery across the range while full PySR mean falls from 59% at $d = 5$ to 41% at $d = 30$; the per-seed value of full search at $d = 20$ varies between 30% and 57%.

safe failure mode.

Limitations

We list the most consequential limitations explicitly, so that downstream users can calibrate their reliance on each headline number.

Sample size of the headline comparison. The headline PySR experiment uses 30 Feynman equations at three seeds. The “60 $\pm$ 0% seed-stable” framing is striking but the absolute statistical power at this sample size is modest—the 99% Wilson interval on 60% from $n = 90$ trials is roughly [46%, 72%]. The 80-equation single-seed scale-up (Supplementary Table 4) is consistent with the headline (66% exact recovery, mean $R^2 = 0.981$) but a full multi-seed Feynman-119 study at this fixed budget remains future work.

Single-seed grid points. The distractor sweep is multi-seed at $d \in \{5, 20, 30\}$ (3 seeds) and $d = 10$ (2 seeds), but the observation-count sweep $q \in \{50, 100\}$ uses a single seed at a higher operating threshold than the deployed one, so it should be read as a qualitative trend (full search worsens with q , DenoisedSR improves) rather than a quantitatively bounded measurement.

The front-end narrows search; it does not generate equations. DenoisedSR is a search-space reducer, not a discovery engine. It cannot reliably assign variables and operators to the correct positions in the expression—that is precisely why we leave final term assignment to a symbolic, verifiable backend. Closing the assignment gap is the central open problem and is not addressed in this work.

Precision on SRSD reflects an aggressive recall-first threshold. On the external SRSD-Feynman suite we report recall 0.967 but precision 0.587: at $\tau = 0.10$ the front-end retains, on average, every dummy column SRSD adds, because SRSD’s dummies are sampled to look physically plausible. A more conservative threshold would raise precision but risks dropping true variables, which is unrecoverable downstream; SRSD-tuning τ per-suite would close part of this gap at the cost of the

Table 1. Unguided search overfits with exotic operators on distractor variables; DenoisedSR recovers the law. For five Feynman laws padded with 20 random distractor columns, full PySR reaches a high (or diverged) held-out R^2 with a physically meaningless expression—variously involving $\cos^2$, $\sinh$, nested $\sin \sin \sinh \tanh \sin$ compositions, fractional powers and distractor columns—whereas DenoisedSR, given the predicted variable support and (where shown) the predicted operator set, recovers an expression structurally matching the true law. Formulas are the actual PySR outputs (literal for full PySR; DenoisedSR expressions are simplified algebraically equivalent rewrites of the literal output where indicated by ✓); x_i denote the (distractor-padded) input columns.

Physical law	Full PySR (all columns/operators)	R^2	DenoisedSR	R^2
Cyclotron freq. $\omega = qvB/p$	$ x_0(x_1+x_2-x_3+\cos(\dots)^2) $ (uses noise col)	0.85	qvB/p ✓	1.000
Doppler shift $\omega = \omega_0/(1-v/c)$	$x_2(\sinh(\dots)+0.79) 1.40$ (not the law)	0.9996	$\omega_0/(1-v/c)$ ✓	1.000
Larmor power $P = \frac{q^2 a^2}{6\pi\epsilon c^3}$	$ 0.48x_3+(x_8^3/x_3^3-\dots/x_{18})^3 $ (diverged)	-2×10^{19}	correct support $\{a, c, \epsilon, q\}$	0.99
Grav. PE diff. $U=Gm_1m_2(\frac{1}{r_2}-\frac{1}{r_1})$	$(x_2-x_3)\sqrt{x_4/\tanh(x_3)-0.56}(\dots)$	0.92	$x_0x_1x_4(x_2-x_3)/(x_2x_3)$ ✓	1.000
Bohr radius $r=4\pi\epsilon\hbar^2/(mq^2)$	$(\dots)[\sin \sin(\dots - \sinh \tanh \sin x_3)]/x_1^2$	0.77	$1.71 x_2^2 x_3/(5.37 x_0 x_1^2)$ ✓	1.000

deployed-threshold universality we currently report.

4 Methods

Problem setup. Each task provides $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$ with $\mathbf{x}_i \in \mathbb{R}^m$, where only an unknown subset $S^* \subseteq \{1, \dots, m\}$ of columns appears in the target law f^* . DenoisedSR predicts $\hat{S} \approx S^*$ (and an optional operator set $\hat{O}$) and passes it to an SR backend run under a fixed time budget. Tasks are padded with random distractor columns to emulate the realistic regime in which a measurement table contains many irrelevant channels.

Front-end architecture. The variable-support predictor is an ensemble of two complementary models. (1) A graph-attention network^{32,33} operates on a graph whose nodes are the input columns; node features combine per-column statistics (moments, correlation with the target, monotonicity and dimensional descriptors) and edges encode a variable co-occurrence prior learned from a knowledge graph of physical laws together with observation-derived correlation. Attention over this graph lets the model reason about which columns *jointly* support a plausible law rather than scoring columns in isolation. (2) A random forest scores each column from the same per-column features. The final score is $s(\text{col}) = \text{RF}(\text{col}) + \sigma(\text{GAT}(\text{col}))$, and a column is retained if $s \geq \tau$ with a low threshold ($\tau = 0.10$) chosen to favour recall. An analogous graph-attention model with graph-level pooling produces the multi-label operator prediction used for the optional operator prior.

Training. The predictors are trained only on internal physical-law records—formula templates with sampled observation tables drawn from a curated knowledge graph of physical laws—never on the external evaluation suites. Training pools of 8000–40 000 formulas were used, with observation tables pre-sampled across variable ranges and a 10–30 distractor curriculum so the front-end is robust to the number of irrelevant columns at test time. An adversarial variant (a generative-adversarial graph-attention model) was explored but gave no consistent gain and is not used.

Backend integration and evaluation. We use PySR⁵ as the primary downstream backend because it exposes variable and operator restrictions directly and is reproducible locally; we additionally verify the gain on gplearn²⁴ (classical genetic programming) to test solver-independence, and the

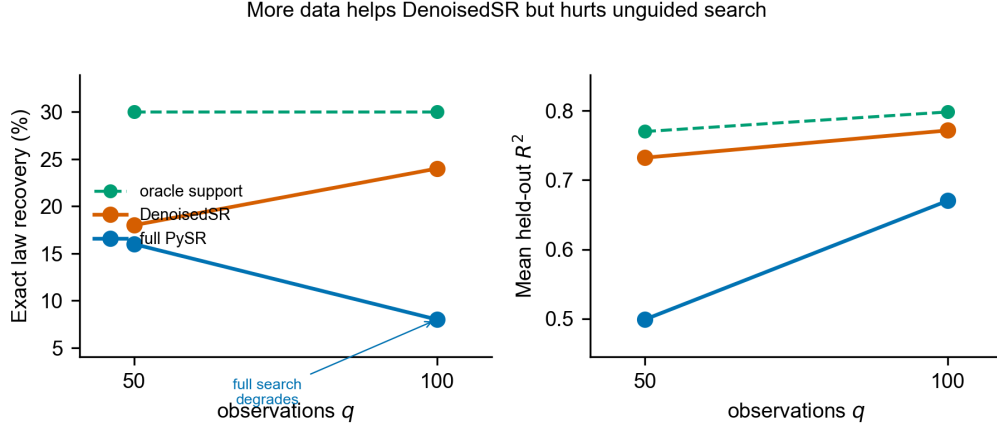


Fig. 7. More data helps DenoisedSR but hurts unguided search. Exact-law recovery (left) and mean held-out R^2 (right) versus the number of observations q , on a 50-equation Feynman set with 10 distractors (single seed, with a front-end variant operating at a higher threshold than the headline $\tau = 0.10$ to expose the qualitative scaling trend). Full search degrades with more data while DenoisedSR improves and approaches the oracle-support upper bound.

same prior interface applies to enumeration¹¹ and generative²² backends. All comparisons fix the backend budget across conditions, so the only changed quantity is the search space exposed to the solver. The headline PySR 30-equation comparison is repeated over three random seeds and reported as mean $\pm$ s.d.; the distractor sweep ($n_{\text{dist}} \in \{5, 10, 20, 30\}$) uses a single seed at each level. Exact recovery denotes held-out $R^2 \geq 0.999$. Support quality is measured by precision and recall against the known law variables, at the deployed ensemble threshold $\tau = 0.10$; the same-size random-support control retains the same number of columns selected without the front-end. Five classical selector baselines (Pearson and Spearman $|\rho|$, mutual information, random-forest importance, and Lasso with cross-validated α) are scored on AI-Feynman at an oracle top- k budget that hands them the true number of variables. Label-noise robustness is evaluated by adding $\eta \cdot \text{s.d.}(y_{\text{clean}})$ Gaussian noise to the target for $\eta \in \{0, 0.01, 0.05, 0.10\}$. Time-to-solution is measured with early stopping on exact recovery, and front-end overhead is measured separately. All reported numbers are produced by the released evaluation scripts from the recorded result artifacts.

Data availability

The external benchmarks (AI-Feynman, PMLB/SRBench, SRSD-Feynman) are public^{2,8,23,41}; the SRSD-Feynman dummy variants used here are obtained from the authors’ Hugging Face repositories. The physical-law training records are materialised from public formula and range information. The recorded result artifacts that back every figure and table are released with the code.

Code availability

Code for the front-end, training, evaluation and figure generation, together with the result artifacts and the trained graph-attention weights, is released at <https://github.com/chenxi-he/sr-frontend> under the MIT License.

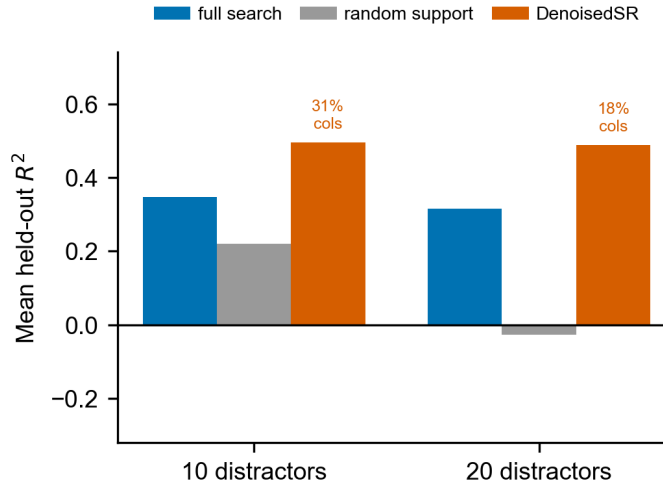


Fig. 8. The gain is search guidance, not fewer columns. Mean held-out R^2 on 24 PMLB/SRBench black-box tasks (three seeds, fixed 5 s budget) at 10 and 20 distractor variables. At matched, reduced column count (labels show columns relative to full search) DenoisedSR improves over full search while a same-size random support degrades far below it.

Author contributions

C.H. conceived the study, designed and implemented the front-end and the evaluation pipeline, performed the experiments and analysis, and wrote the manuscript. J.C. supervised the project, advised on the methodology and the physics evaluation, and revised the manuscript. Both authors approved the final version.

Competing interests

The authors declare no competing interests.

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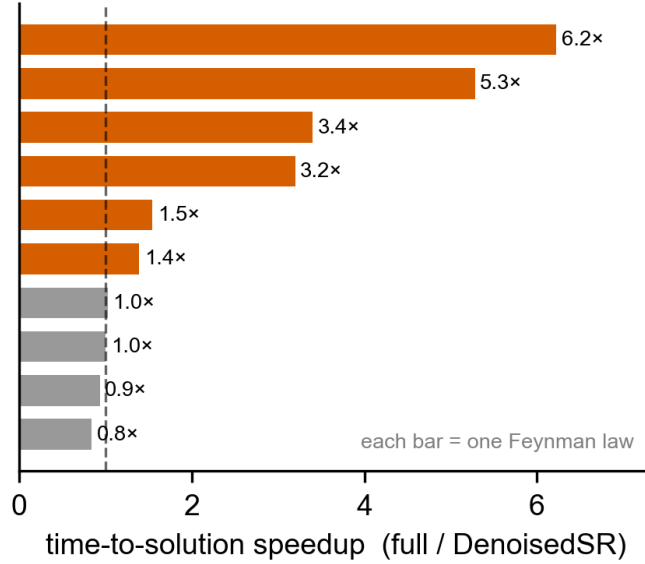


Fig. 9. Pruning accelerates time-to-solution. Per-formula speedup (full-search time divided by DenoisedSR time) with early stopping on exact recovery, on 10 Feynman formulas at a single seed. Non-trivial physics formulas solve up to $6\times$ faster (median $\sim 1.5\times$); deep-tree formulas (depth-, not variable-limited) show little gain.

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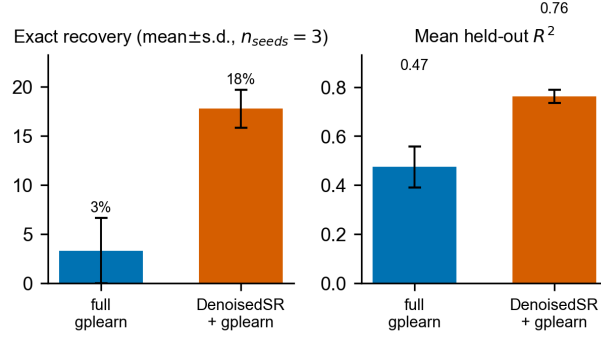


Fig. 10. Same gain transfers to a second backend. Exact recovery and mean held-out R^2 on the 30-equation Feynman subset with the gplearn backend²⁴ (20 generations, population 500, ~ 20 distractors; mean $\pm$ s.d. over three random seeds). Full gplearn rarely recovers a law exactly ($3.3 \pm 3.3\%$); supplied with the DenoisedSR variable prior over the same 3.4 predicted columns it recovers $17.8 \pm 1.9\%$ exact, raising mean held-out R^2 from 0.48 to 0.76.

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Supplementary information

Table 2. Summary of variable-support recovery across selectors and suites. Variable-support recall, with the fraction of tasks on which the selector attains *perfect* recall (every true variable retained). All selectors except DenoisedSR are run at an oracle top- k budget that hands them the true variable count for free. Recall on AI-Feynman is also reported under relative Gaussian label noise η ; entries marked — were not part of the noise sweep.

Selector	AI-Feynman (118)		SRSD-Feynman (120)	AI-Feynman noise (η)		
	recall	perfect %	recall	$\eta=0.01$	$\eta=0.05$	$\eta=0.10$
Pearson $ \rho $	0.768	45.8	—	—	—	—
Spearman $ \rho $	0.776	45.8	—	—	—	—
mutual information	0.634	20.3	—	—	—	—
random-forest importance	0.812	44.9	—	—	—	—
Lasso (CV) <i>oracle k</i>	0.905	71.2	0.723	0.904	0.904	0.886
DenoisedSR (ours, $\tau=0.10$)	1.000	100.0	0.967	1.000	1.000	1.000

Table 3. Backend integration summary. Mean exact-law recovery and mean held-out R^2 on the 30-equation Feynman headline subset ($n_{\text{dist}} = 20$, $q = 100$). Errors are $\pm$ s.d. over three random seeds. The variable prior shrinks the candidate-column set from 23.2 to 3.4 in both backends.

Condition	Exact recovery (%)	Mean held-out R^2
<i>PySR (10 s budget, full 17-op library)</i>		
full search	42.2 ± 13.5	0.906 ± 0.045
+ DenoisedSR variable prior	60.0 ± 0.0	0.980 ± 0.011
+ DenoisedSR variable + operator prior	60.0 ± 11.5	0.979 ± 0.011
<i>gplearn (20 generations, population 500, 10-op library)</i>		
full search	3.3 ± 3.3	0.475 ± 0.083
+ DenoisedSR variable prior	17.8 ± 1.9	0.763 ± 0.029

Table 4. Headline result scales to 80 Feynman equations. Repeating the PySR comparison at the same configuration ($n_{\text{dist}} = 20$, $q = 100$, $\tau = 0.10$, single seed) on the first 80 Feynman tasks (chosen by $2 \leq |\text{features}| \leq 9$) rather than the headline 30. The gain is preserved at larger scale; full PySR’s higher absolute rate at this scale (54% vs. 30–57% across seeds at $n=30$) is consistent with the 30-equation subset including a higher fraction of harder formulas.

Condition	Exact (%)	Mean held-out R^2	Variable recall
full PySR (all 23 cols, 17 ops)	54	0.900	—
+ DenoisedSR variable prior (3.4 cols)	66	0.981	1.000
+ DenoisedSR variable + operator prior	68	0.981	1.000

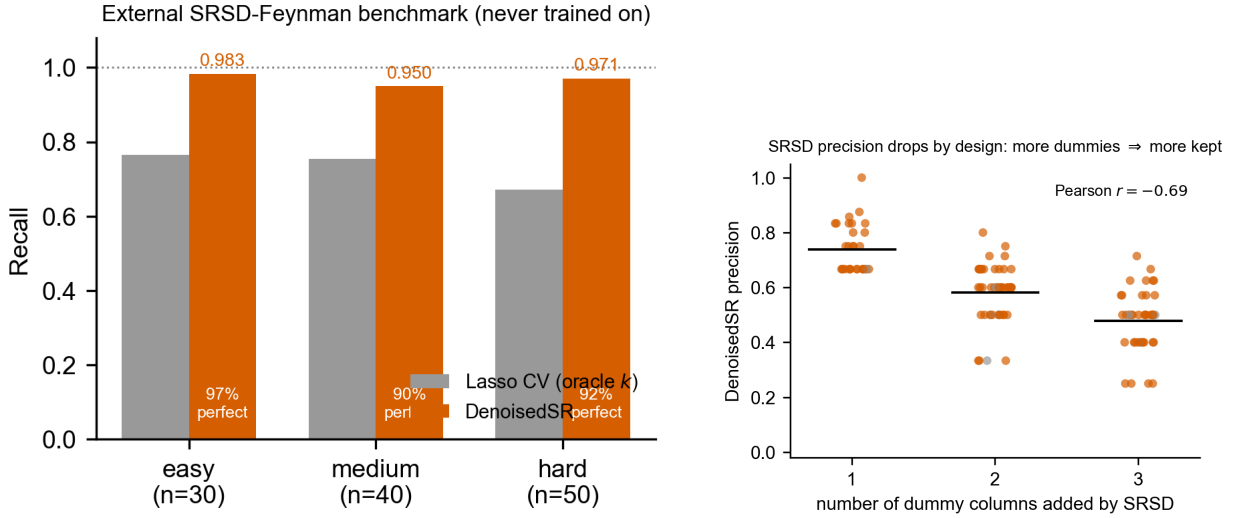


Fig. 11. Detail on SRSD-Feynman. *Left:* recall on each SRSD difficulty split (30 easy, 40 medium, 50 hard) for DenoisedSR vs. Lasso at oracle top- k ; DenoisedSR exceeds 0.95 recall on every split. *Right:* diagnostic of the apparent precision drop on SRSD (0.59 overall). Each marker is one of the 120 SRSD tasks (orange = perfect recall, grey = imperfect recall); horizontal bars show the binned mean precision. Precision falls monotonically with the number of dummy columns SRSD adds to the task (Pearson $r = -0.69$) because the front-end at $\tau=0.10$ retains physically plausible dummies rather than discarding them; on tasks with no dummies our precision is near 1.